

Cloud Computing Seminar

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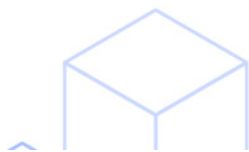
Term: Spring 2025

1. Set up an AWS Free Tier Account:
 - If you haven't done so, create an AWS Free Tier account at AWS Sign-Up.
 - Use an email not associated with another AWS account.



Explore Free Tier products with a new AWS account.

To learn more, visit aws.amazon.com/free.



Sign up for AWS

Create your password

✔ It's you! Your email address has been successfully verified. ✕

Your password provides you with sign in access to AWS, so it's important we get it right.

Root user password

.....

Confirm root user password

.....

Continue (step 1 of 5)

OR

Sign in to an existing AWS account

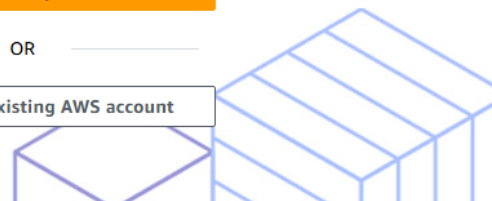


Fig 1.1 Sign up Root user

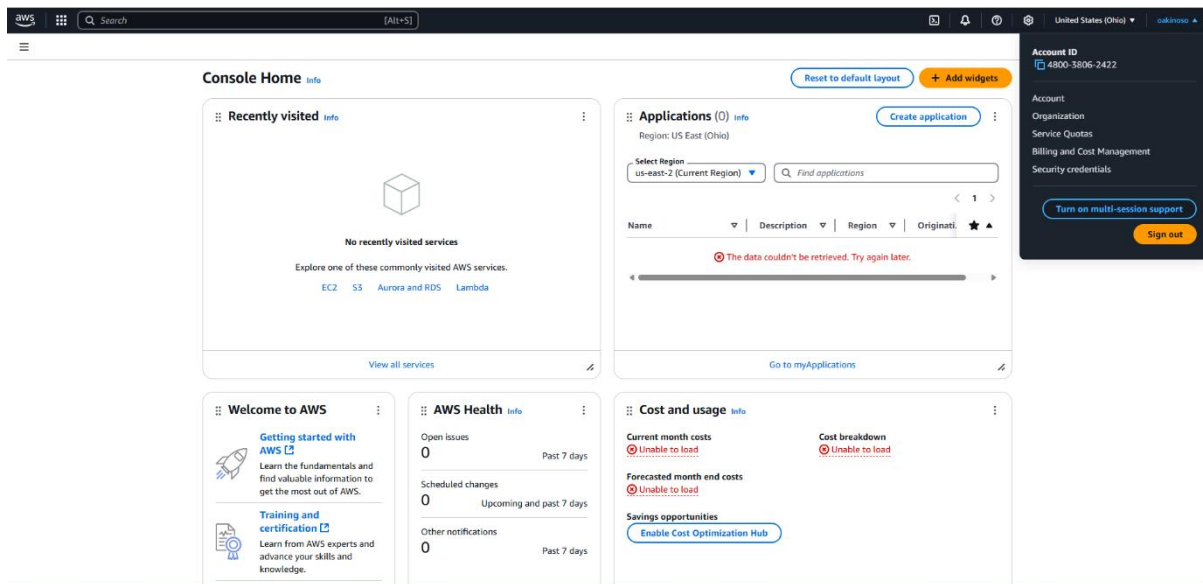


Fig. 1.2 Creation of root user

2. Add MFA for the root account:

Go to: AWS Console → IAM → Security Credentials.

Enable MFA: Choose "Virtual MFA" and use Google Authenticator/Authy.

Take a screenshot of the MFA setup confirmation.



Keep your account secure



Registering your MFA for root credentials will be required in 35 days.

Improve the security of your account by registering **multi-factor authentication (MFA)** using one of the options below. This provides a second means of verifying your identity in addition to your password. [Learn more](#)

MFA device name

Device Name

Maximum 128 characters. Use alphanumeric and '+ = , . @ - _' characters.



Passkey or Security key

Authenticate using your fingerprint, face, or screen lock. Create a passkey on this device or use another device, like a FIDO2 security key.



Authenticator app

Authenticate using a code generated by an app installed on your mobile device or computer.



Hardware TOTP Token

Authenticate using a code generated by hardware TOTP token or other hardware devices.

Skip for now

Next

Fig 2.1 MFA Initiation



Set up authenticator app

An authenticator app is an application running on your device that you can configure by scanning a QR code.

- 1 Install a compatible application such as Google Authenticator, Duo Mobile, or Authy app on your mobile device or computer.

[See a list of compatible applications](#) 

- 2



Open your authenticator app, choose Show QR code on this page, then use the app to scan the code.

Alternatively, you can type a secret key.

[\(Show secret key\)](#)

- 3

Enter two consecutive codes from your authenticator app. Enter the first code, wait a few seconds and then enter the second.

First code

Second code

Cancel

Register MFA

Fig 2.2 MFA using Microsoft Authenticator

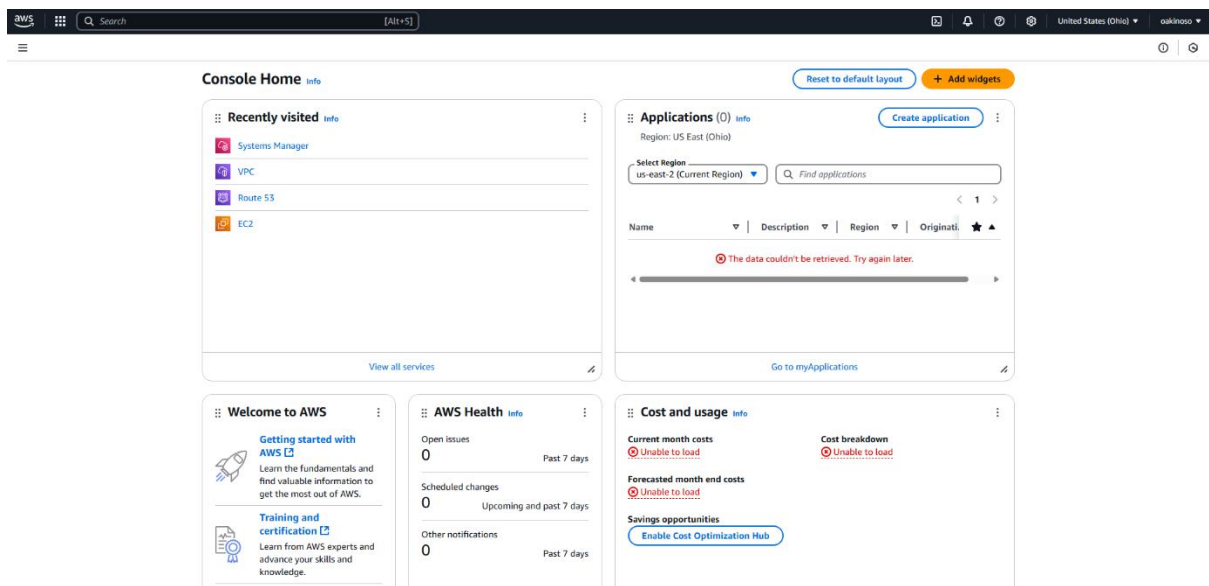


Fig 2.3 Root User Creation Successful

3. Create another account and work in this account from now on:
 - Go to: IAM → Users → Add User.
 - Username: your-name-id.
 - Attach Policy: AdministratorAccess.
 - Set Console Access: Enable sign-in for this user.
 - Log out from the root account and only use this IAM user from now on.
 - Take a screenshot of the IAM user list.

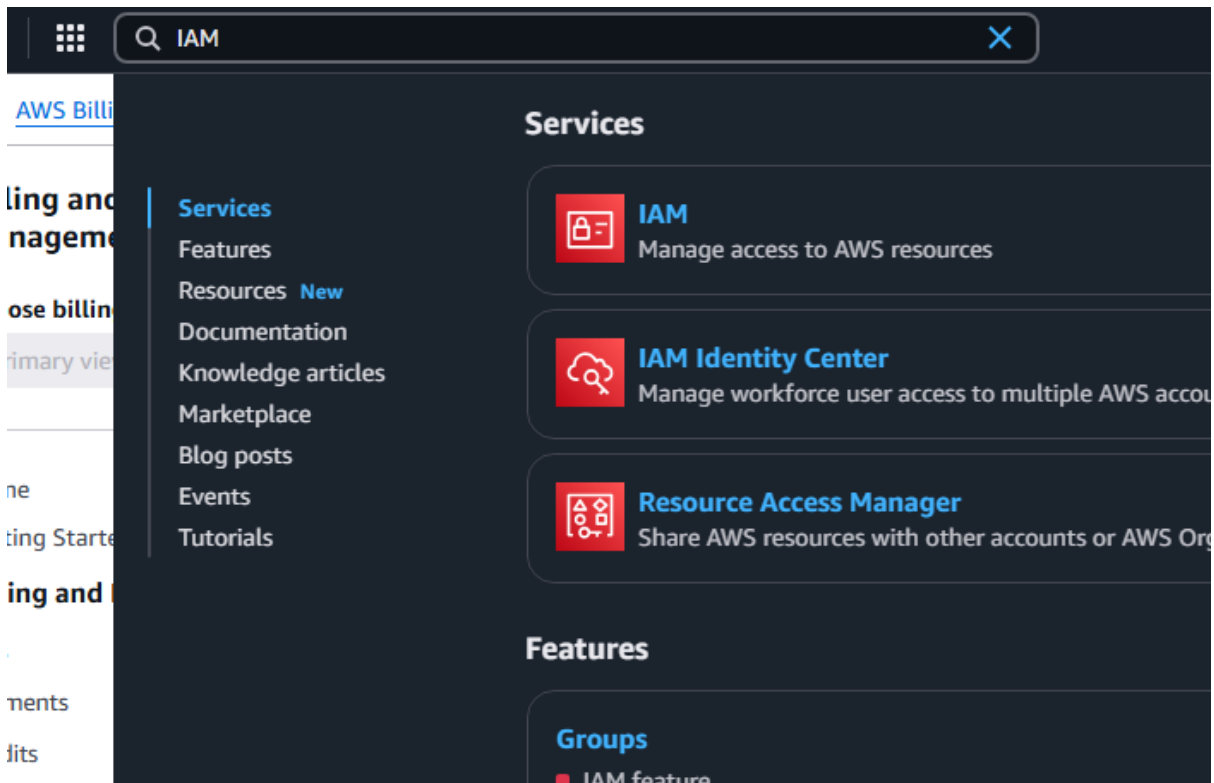


Fig 3.1 IAM User Creation

Set permissions

Add user to an existing group or create a new one. Using groups is a best-practice way to manage user's permissions by job functions. [Learn more](#)

Permissions options

☐ Add user to group

Add user to an existing group, or create a new group. We recommend using groups to manage user permissions by job function.

☐ Copy permissions

Copy all group memberships, attached managed policies, and inline policies from an existing user.

☒ Attach policies directly

Attach a managed policy directly to a user. As a best practice, we recommend attaching policies to a group instead. Then, add the user to the appropriate group.

Permissions policies (1/1346)

Choose one or more policies to attach to your new user.



Create policy

Search

Filter by Type

All types

< 1 2 3 4 5 6 7 ... 68 >

☐	Policy name	Type	Attached entities
☐	AccessAnalyzerServiceRolePolicy	AWS managed	0
☒	AdministratorAccess	AWS managed - job function	1
☐	AdministratorAccess-Amplify	AWS managed	0
☐	AdministratorAccess-AWSElasticBeanstalk	AWS managed	0
☐	AIOpsAssistantPolicy	AWS managed	0
☐	AIOpsConsoleAdminPolicy	AWS managed	0
☐	AIOpsOperatorAccess	AWS managed	0

Fig 3.2 IAM Attach Policy

Review and create

Review your choices. After you create the user, you can view and download the autogenerated password, if enabled.

User details

User name
olamide_miu

Console password type
None

Require password reset
No

Permissions summary

Name 



Type



Used as



[AdministratorAccess](#)

AWS managed - job function

Permissions policy

Tags - optional

Tags are key-value pairs you can add to AWS resources to help identify, organize, or search for resources. Choose any tags you want to associate with this user.

No tags associated with the resource.

[Add new tag](#)

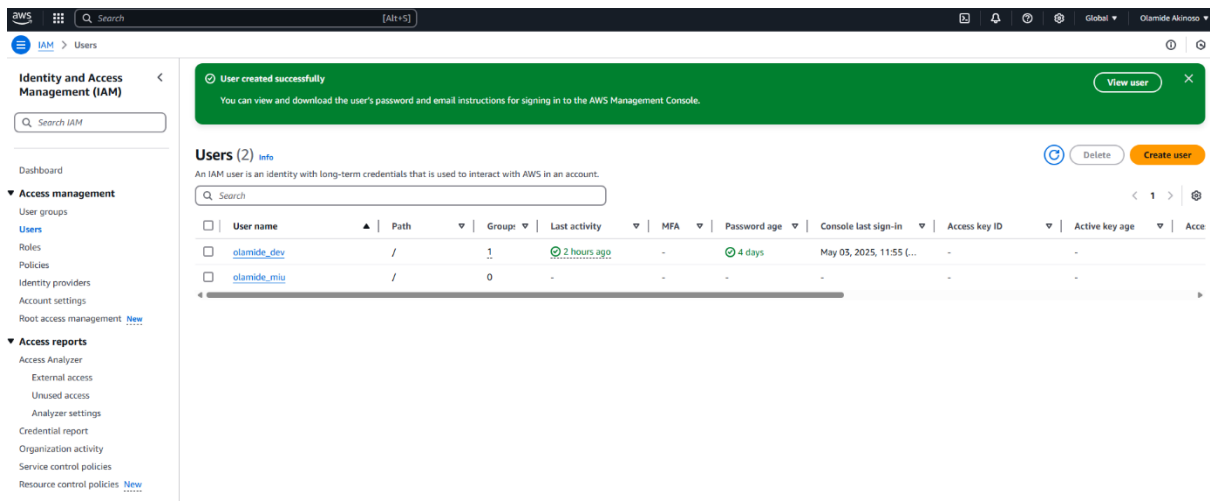
You can add up to 50 more tags.

[Cancel](#)

[Previous](#)

[Create user](#)

Fig 3.3 Review IAM User Creation



Identity and Access Management (IAM)

Search IAM

Dashboard

Access management

- User groups
- Users**
- Roles
- Policies
- Identity providers
- Account settings
- Root access management [New](#)

Access reports

- Access Analyzer
- External access
- Unused access
- Analyzer settings
- Credential report
- Organization activity
- Service control policies
- Resource control policies [New](#)

Users (2) [info](#)

An IAM user is an identity with long-term credentials that is used to interact with AWS in an account.

Search

☐	User name	Path	Groups	Last activity	MFA	Password age	Console last sign-in	Access key ID	Active key age	Access
☐	olamide_dev	/	1	2 hours ago	-	4 days	May 05, 2025, 11:55 [...]	-	-	
☐	olamide_miu	/	0	-	-	-	-	-	-	

Fig 3.4 IAM User lists

4. Use your root account to create Billing Alert:
 - Go to: Billing Dashboard → Preferences.
 - Enable billing alerts and set up an email notification.
 - Take a screenshot of the billing alert settings.

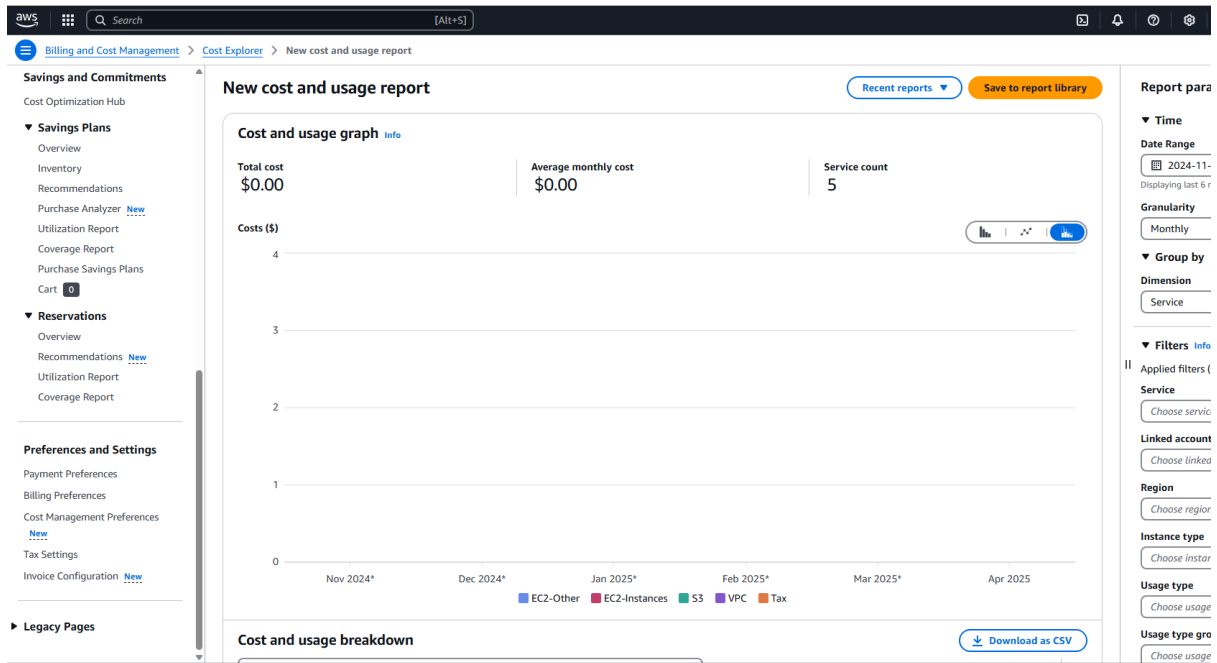


Fig 4.1 Billing Dashboard

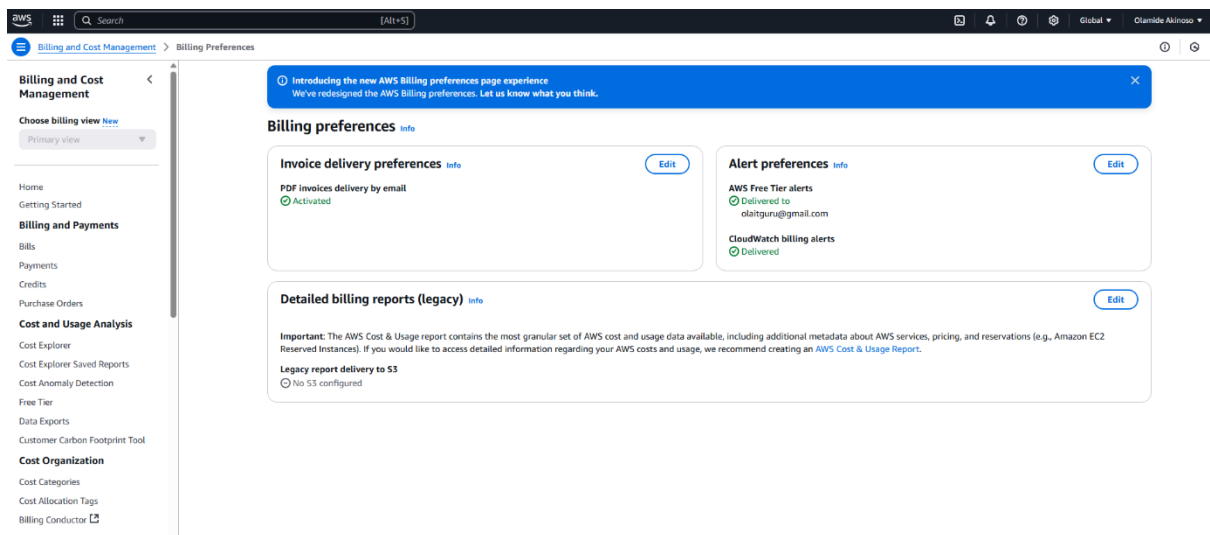


Fig 4.2 Enabling Alerts

5. Use your root account to create Budget Alert:

- Go to: AWS Budgets → Create Budget.
- Type: Cost Budget.
- Threshold: \$1 (or Free Tier limit).
- Notification: Send email alerts when spending approaches the limit.
- Take a screenshot of the budget alert setup.

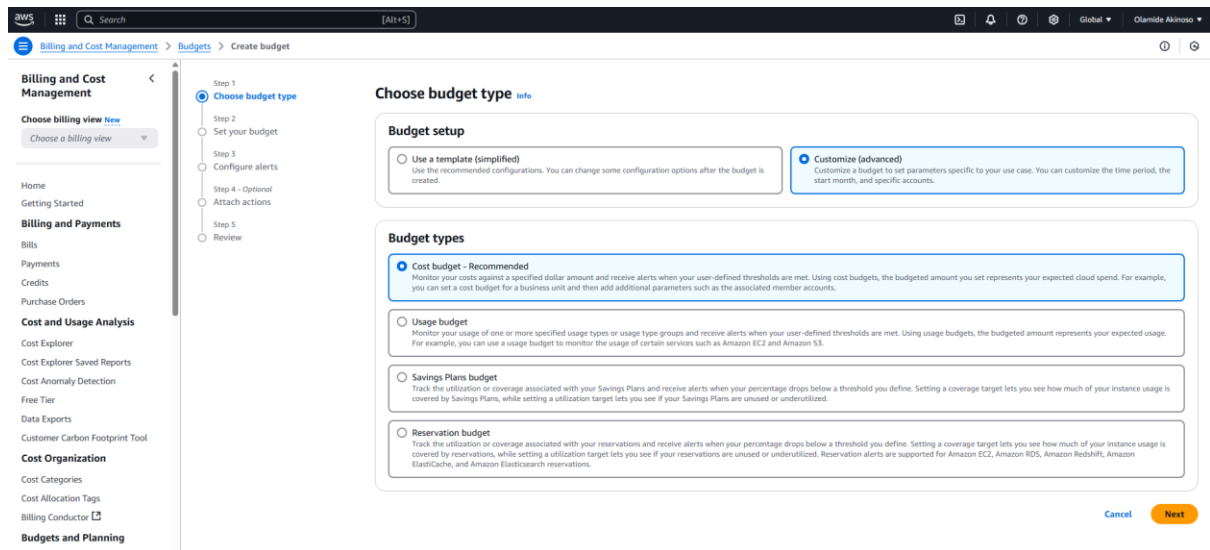


Fig 5.1 AWS Budget Creation

Your budgeted amount: **\$1.00**
To change your budgeted amount, go back to step 2.

▼ Alert #1

Remove

Set alert threshold

Threshold

When should this alert be triggered?

80

% of budgeted amount ▼

Trigger

How should this alert be triggered?

Actual ▼

Notification preferences

Select one or more notification preferences to receive alerts.

Email recipients

Specify the email recipients you want to notify when the threshold has exceeded.

oakinoso@miu.edu

Maximum number of email recipients is 10.

► Amazon SNS Alerts - *Optional Info*

► AWS Chatbot Alerts

+ Add alert threshold

Cancel

Previous

Next

Budget preview

Cost Data

May 2024 - May 2025 (

1

0.8

0.6

0.4

0.2

0

May 2024

Jul

Actual cost

== E

View in AWS Co

Alerts

No alerts configu

Fig 5.2 Budget Alerts and Notification

Attach actions - *Optional* Info

▼ Using budgets actions



What is a budget action?

A budget action allows you to define and trigger cost saving responses to reinforce a cost-conscious culture. You have the option to attach actions that run whenever your alert threshold has been exceeded, such as stopping an EC2 instance from incurring any further costs. You can select the alerts to which you would like to attach actions, then define these actions.



How to get started?

To create a budget action, you will first need an alert threshold created from step 2. If you have already created an alert threshold select the type of action you want.

▼ Alert #1 (0 actions attached)

Threshold
80%
Threshold measured against
Actual Costs

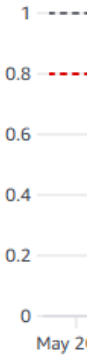
[Add action](#)

Email recipients
oakinoso@miu.edu
Amazon SNS
Not configured

[Cancel](#) [Previous](#) [Next](#)

Budget

Cost Data
May 2024



■ Actual

[View](#)

Alerts

► Actual

Fig 5.3 Budget Threshold and Alert Settings

Your budget **miu_test** has been created successfully. [Submit feedback](#)

Budgets (2) Info

Type - Show all budgets ▼

[Download CSV](#) [Actions](#) [Create budget](#)

☐	Name	▲	Thresholds	▼	Budget	Amount used	Forecasted amount	Current vs. budgeted	Forecasted vs. budgeted
☐	miu_test		OK		\$1.00	-	-	0.00%	-
☐	My Monthly Cost Budget		OK		\$10.00	\$0.00	-	0.00%	-

Fig 5.4 Budget Alert setup

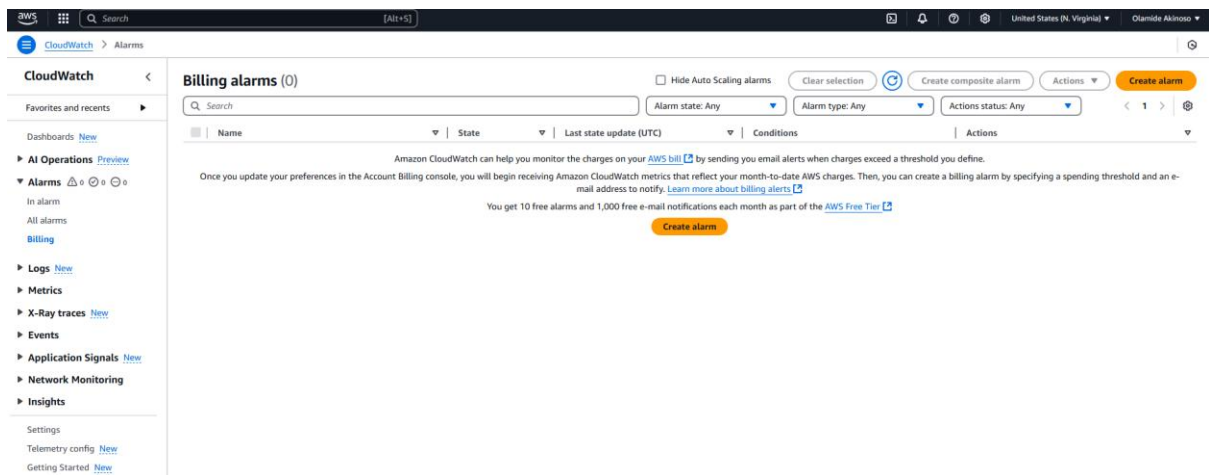


Fig 5.5 Billing Alarm using Cloudwatch

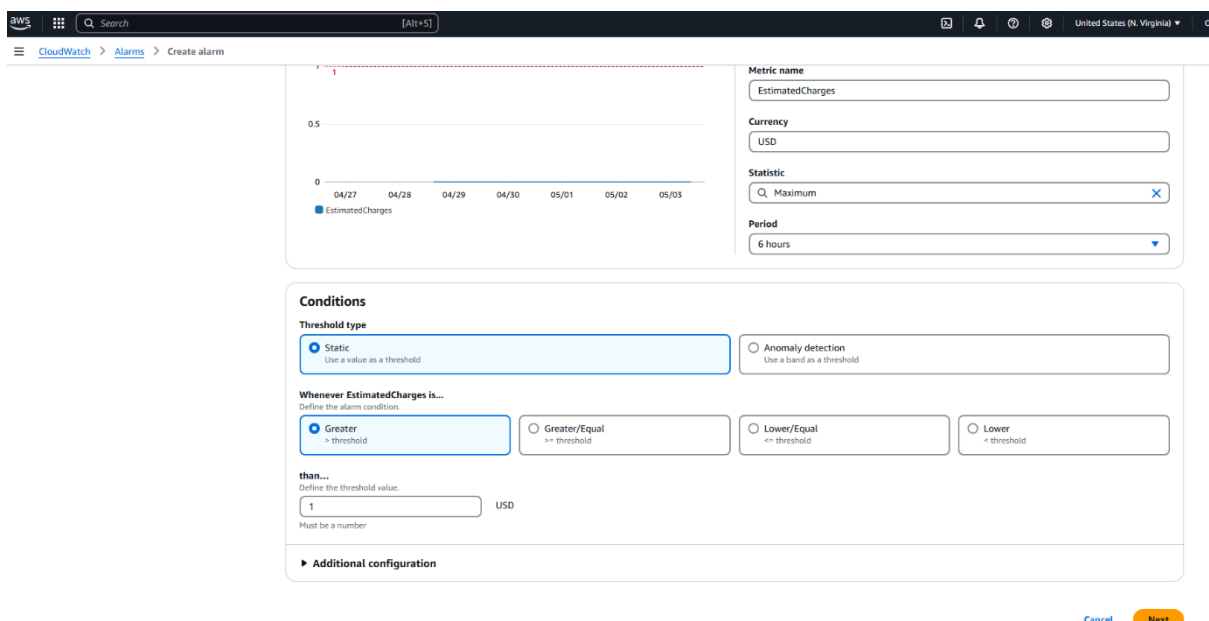


Fig 5.6 Billing Alarm using Cloudwatch

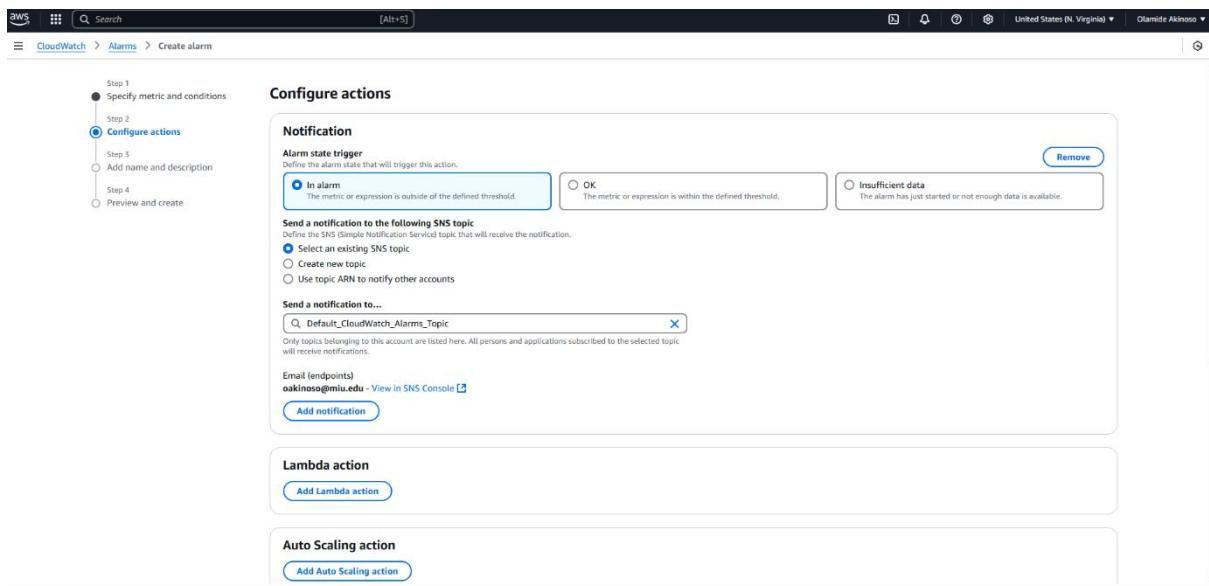


Fig 5.7 Billing Alarm threshold using Cloudwatch

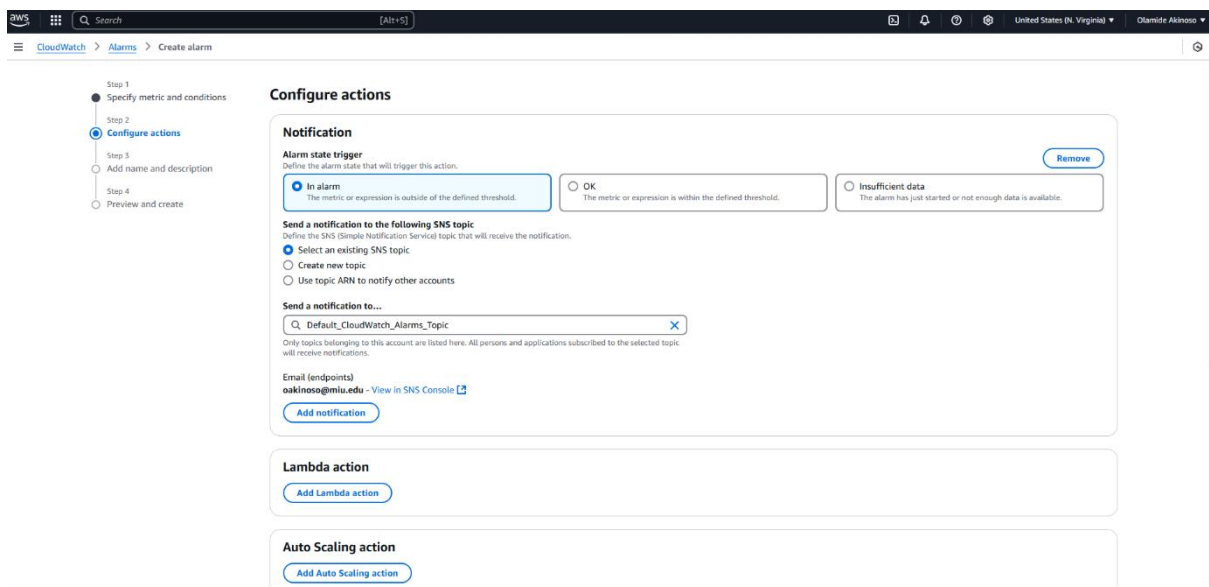


Fig 5.8 Create Topic Billing Alarm

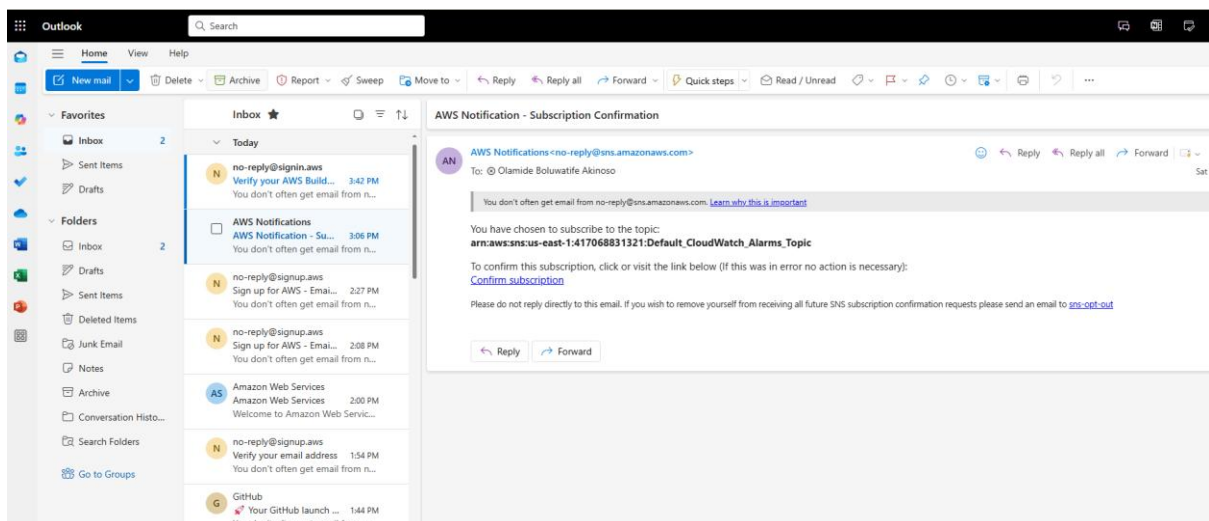


Fig 5.9 Notification via Email